

Linda Bushel pool deck — material take-off J1

Issue: Preliminary Construction and Supplier Pricing Basis - Subject to Permit Review and Field Verification

Date: 2026-10-01

This take-off reconciles to drawings/2026-10-01-j1-preliminary-11x17.pdf.

Issue J1 uses 15 piers, 16 joist lines, and 10 stringers.

Issue P1 and the capability-test sheets remain historical.

It does not contain a Brayman cost, a customer price, or a gross margin.

No internal estimate and no customer estimate are part of this issue.

Waste is the offcut from the stated stock. No extra percentage is applied.

Offcuts are not deducted from the purchase count.

Geometry used

Item	Value	Source
Pool radius	10.500 ft	Recorded 21 ft diameter
Front edge	Y = 15.500 ft	Radius plus recorded 5 ft clear depth
Side edges	X = ± 9.000 ft	Recorded 18 ft width
Side intersection	Y = 5.408 ft	Circle at X = ± 9
Upper deck area	116.801 sq ft	18 ft rectangle 181.650 minus pool segment 64.849
Pool-edge arc	21.624 ft	Recorded radius between the side intersections
Main beam	Y = 13.000 ft, X = ± 6.000 ft	Preliminary Option A layout
Wing end piers	X = ± 9.0 ft, Y = 8.5 ft	Preliminary Option A layout
Header segment	8 $\times$ 2.695 ft	Outside the pool; sagitta 1.04 in
Stair run	3.667 ft	4 $\times$ 11 in; tread count is a landing assumption
Lower deck Y	19.167 to 22.167 ft	After the stair run; recorded 3 ft depth

Lower-deck height

Intended finished walking surface 12 in. Dressed stack $1.00 + 7.25 + 9.25 = 17.50$ in. Difference 5.50 in. Post cut length is a field reconciliation before construction. Lower posts are ordered uncut.

Upper joist cut list

Stock is $2 \times 8 \times 12$ ft. A pair is accepted only when both cuts plus a $1/8$ in kerf fit on the stick.

Stick	Cuts (ft)	Offcut (ft)
J1	10.092	1.908
J2	10.092	1.908
J3	9.572	2.428
J4	9.572	2.428
J5	7.985	4.015
J6	7.985	4.015
J7	$6.883 + 5.021$	0.096
J8	$6.883 + 5.021$	0.096
J9	$6.094 + 5.192$	0.714
J10	$6.094 + 5.192$	0.714
J11	$5.543 + 5.543$	0.914

Sixteen joists. Net lineal 112.766 ft. Purchased 11×12 ft = 132 ft.

Joist positions from the centreline, with length to the pool arc:

X	Length
-9.000 ft	10.092 ft
-8.667 ft	9.572 ft
-7.333 ft	7.985 ft
-6.000 ft	6.883 ft
-4.667 ft	6.094 ft
-3.333 ft	5.543 ft
-2.000 ft	5.192 ft
-0.667 ft	5.021 ft
+0.667 ft	5.021 ft
+2.000 ft	5.192 ft
+3.333 ft	5.543 ft
+4.667 ft	6.094 ft
+6.000 ft	6.883 ft
+7.333 ft	7.985 ft
+8.667 ft	9.572 ft
+9.000 ft	10.092 ft

Other 2×8 × 12 ft sticks

Use	Sticks	Allocation
Segmented header	2	4 segments of 2.695 ft on each stick
Front and main-beam blocking	3	22 pieces × 14.5 in, plus two 2.5 in closers
Wing blocking	1	4 blocks along the wings
Lower-deck joists	3	Nine 3 ft joists, three per stick after kerf
Total 2×8 × 12 ft with the upper joists	20	Offcuts not reused

5/4 × 6 cut list

Upper full-width boards: 11 pieces at 18 ft, each cut from a 20 ft board. Offcut 2 ft each. One of those boards is ripped to 5.0 in width.

Curve-side piece lengths, each bought twice (left and right), measured at the wide edge of the row:

- Row 1: 9.000 ft each side
- Row 2: 5.932 ft each side
- Row 3: 4.709 ft each side
- Row 4: 3.805 ft each side
- Row 5: 3.072 ft each side
- Row 6: 2.452 ft each side
- Row 7: 1.916 ft each side
- Row 8: 1.445 ft each side
- Row 9: 1.028 ft each side
- Row 10: 0.657 ft each side
- Row 11: 0.326 ft each side

Corner remainder below the last row: 0.60 in. No extra board.

Stock length	Cuts placed on that stick (ft)	Offcut (ft)
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	10.000	0.000
10 ft	9.000 + 0.657 + 0.326	0.017
10 ft	9.000 + 0.657 + 0.326	0.017
8 ft	5.932 + 1.916	0.153
8 ft	5.932 + 1.916	0.153
8 ft	4.709 + 3.072	0.218
8 ft	4.709 + 3.072	0.218
8 ft	3.805 + 3.805	0.389
8 ft	2.452 + 2.452 + 1.445 + 1.445	0.206
8 ft	1.028 + 1.028	5.944

Separate from that pack: 11 pieces of 5/4×6 × 20 ft for the full-width upper rows.

Guard runs

Run	Length (ft)	Treatment	Kits
Upper left side	10.092	straight kit run	2
Upper right side	10.092	straight kit run	2
Upper front left of stair	4.000	straight kit run	1
Upper front right of stair	4.000	straight kit run	1
Lower left side	3.000	straight kit run	1
Lower right side	3.000	straight kit run	1
Lower front left of gate	3.250	straight kit run	1
Lower front right of gate	3.250	straight kit run	1
Stair left	4.845	custom sloped match	custom, not a 6 ft kit
Stair right	4.845	custom sloped match	custom, not a 6 ft kit

Fastener counts tied to the drawing

Item	Count	Basis
Helical piers	15	One at each drawn location. Not a sonotube.
Helical-pier brackets	15	One per pier unless the pier includes the bracket
Post-to-beam connectors	15	One per post
Header connectors	16	One per upper joist

Item	Count	Basis
Stringer connectors	10	One per stringer
Beam stitch bolts	76	Two rows at 16 in OC between posts; pricing allowance
Nuts / washers	76 / 152	For those bolts
Structural screws	100	Two at each joist bearing on a beam or wing
Deck screws	798	Two per board crossing. Upper full-width 352, curve sides 160, lower deck 126, treads 160

Guard-post blocking, hinge count, bolt diameter, and connector brand are not quantified beyond the flags in the supplier request.

Helical piers

15 helical piers, one at each drawn location. Sonotubes and concrete are not in this issue. Shaft, helix, length, torque, and capacity are not specified and are not certified.

Category waste

Category	What the waste is
Helical piers	No waste percent. Shaft, helix, and length are not specified.
Posts and brackets	A paired upper stick has about 31 in left after two 32.5 in cuts. The ninth post uses one stick. Do not pre-cut.
Beams	Front offcut none on a 6 ft + 6 ft cut. Wing offcut about 1.18 ft × 2. Main and lower offcut none.
Joists, header, blocking	Upper joist stock 132 ft against net 112.766 ft. Header, blocking, and lower offcuts are on the cut list below. Offcuts are not reused to reduce the purchase.
Stringers	About 35 in left on each 8 ft piece after a 3 in end-cut allowance. Plumb-cut drawing is not included.
Decking and treads	2 ft off each 20 ft upper board, plus bin-pack offcut on the shorter pieces.
Skirting	Seven 12 in cuts per stick with 1/8 in kerf leaves about 11 in. Five sticks.
Rails and guard posts	Each straight segment starts a new kit. Short segments waste the unused portion of the 6 ft module. Posts are not pre-cut because the below-deck length is not issued.
Gate	None calculated.
Fasteners	No extra percent.

Calculation file: takeoff/issue_j1.py.